Hugo Marquerie 11/01/2026

Teo aproximacion identidad convolucion

Teorema 1. *Sea $\phi \in \mathcal{L}^1(\mathbb{R}^n)$ tal que $\phi \geqslant 0$ y $\int_{\mathbb{R}^n} \phi(x)\,\mathrm{d}x = 1$. Entonces,*

(1) $\forall f \in \mathcal{L}^p(\mathbb{R}^n)$ *con* $p \in [1,\infty)$: $\lim_{t\to 0^+} \|f * \phi_t - f\|_p = 0$.

(2) $\forall f \in \mathcal{L}^\infty(\mathbb{R}^n)$ *uniformemente continua* : $\lim_{t\to 0^+} \|f * \phi_t - f\|_\infty = 0$.

(3) $\forall f$ *continua en* $x_0 \in \mathbb{R}^n$: $\lim_{t\to 0^+} f * \phi_t(x_0) = f(x_0)$.

***Demostración*:** Como $\int_{\mathbb{R}^n} \phi(x)\,\mathrm{d}x = 1$, por la `obs-propiedades-dilatacion-isotropica/`Observació sabemos que $\forall t > 0 : \int_{\mathbb{R}^n} \phi_t(x)\,\mathrm{d}x = 1$. Entonces, $\forall x \in \mathbb{R}^n$:

$$\begin{aligned} |f * \phi_t(x) - f(x)| &= \left| \int_{\mathbb{R}^n} f(x-y)\,\phi_t(y)\,\mathrm{d}y - \int_{\mathbb{R}^n} f(x)\,\phi_t(y)\,\mathrm{d}y \right| \qquad (1) \\ &= \left| \int_{\mathbb{R}^n} [f(x-y) - f(x)]\;\phi_t(y)\,\mathrm{d}y \right| \leqslant \int_{\mathbb{R}^n} |T_y f(x) - f(x)|\;\phi_t(y)\,\mathrm{d}y. \end{aligned}$$

(1) Sea q el exponente conjugado de p (i.e. $1/p + 1/q = 1$). Entonces, por Hölder,

$$\begin{aligned} |f * \phi_t(x) - f(x)| &\overset{\overset{(??)}{\downarrow}}{\leqslant} \int_{\mathbb{R}^n} |T_y f(x) - f(x)|\; \phi_t^{1/p}(y)\,\phi_t^{1/q}(y)\,\mathrm{d}y \\ &\overset{\overset{\text{Hölder}}{\downarrow}}{\leqslant} \left(\int_{\mathbb{R}^n} |T_y f(x) - f(x)|^p\; \phi_t(y)\,\mathrm{d}y \right)^{1/p} \cdot \underbrace{\left(\int_{\mathbb{R}^n} \phi_t(y)\,\mathrm{d}y \right)^{1/q}}_{1}. \qquad (2) \end{aligned}$$

$$\begin{aligned} \|f * \phi_t - f\|_p^p = \int_{\mathbb{R}^n} |f * \phi_t(x) - f(x)|^p\,\mathrm{d}x &\overset{\overset{(??)}{\downarrow}}{\leqslant} \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} |T_y f(x) - f(x)|^p\; \phi_t(y)\,\mathrm{d}y\,\mathrm{d}x = \\ \overset{\overset{\text{Fubini}}{\downarrow}}{\leqslant} \int_{\mathbb{R}^n} \phi_t(y) \left(\int_{\mathbb{R}^n} |T_y f(x) - f(x)|^p\,\mathrm{d}x \right) \mathrm{d}y &= \int_{\mathbb{R}^n} \phi_t(y)\; \|T_y f - f\|_p^p\,\mathrm{d}y. \qquad (3) \end{aligned}$$

Por el `lem-convergencia-lp-traslacion/`Lema 1, dado $\varepsilon > 0$, existe $\eta > 0$ tal que $\forall y \in \mathbb{R}^n : \|y\| < \eta : \|T_y f - f\|_p < \varepsilon$. Entonces, sea $\eta > 0$ tal que $\|y\| < \eta \implies \|T_y f - f\|_p \leqslant \frac{\varepsilon}{2^{1/p}}$. Por la `obs-propiedades-dilatacion-isotropica/`Observación 1,

$\exists \delta > 0 : \forall t \in (0,\delta) : \int_{\|y\|\geqslant \eta} \phi_t(y)\,\mathrm{d}y < \frac{\varepsilon^p}{2^{p+1}\|f\|_p^p}$. Entonces, para $t \in (0,\delta)$,

$$\begin{aligned}\|f * \phi_t - f\|_p^p &\overset{\substack{(\textbf{??})\\ \downarrow}}{\leqslant} \int_{\|y\|<\eta} \|T_y f - f\|_p^p\, \phi_t(y)\,\mathrm{d}y + \int_{\|y\|\geqslant\eta} \|T_y f - f\|_p^p\, \phi_t(y)\,\mathrm{d}y \\ &\leqslant \frac{\varepsilon^p}{2}\int_{\|y\|\leqslant\eta} \phi_t(y)\,\mathrm{d}y + 2^p\,\|f\|_p^p\,\frac{\varepsilon^p}{2^{p+1}\,\|f\|_p^p} \leqslant \frac{\varepsilon^p}{2} + \frac{\varepsilon^p}{2} = \varepsilon^p.\end{aligned}$$

Por tanto, $\lim_{t\to 0^+} \|f * \phi_t - f\|_p = 0$.

(2) Como f es uniformemente continua, $\forall\varepsilon > 0 : \exists\eta > 0 : \forall x, y \in \mathbb{R}^n : \|y\| < \eta \implies |f(x-y) - f(x)| < \frac{\varepsilon}{2}$. Entonces, por (**??**), tenemos que

$$\begin{aligned}|f * \phi_t(x) - f(x)| &\overset{\substack{(\textbf{??})\\ \downarrow}}{\leqslant} \int_{\|y\|<\eta} |T_y f(x) - f(x)|\, \phi_t(y)\,\mathrm{d}y + \int_{\|y\|\geqslant\eta} |T_y f(x) - f(x)|\, \phi_t(y)\,\mathrm{d}y \\ &\leqslant \frac{\varepsilon}{2}\int_{\mathbb{R}^n} \phi_t(y)\,\mathrm{d}y + 2\,\|f\|_\infty \int_{\|y\|\geqslant\eta} \phi_t(y)\,\mathrm{d}y \leqslant \frac{\varepsilon}{2} + 2\,\|f\|_\infty\,\frac{\varepsilon}{4\,\|f\|_\infty} = \varepsilon.\end{aligned}$$

Por tanto, $\lim_{t\to 0^+} \|f * \phi_t - f\|_\infty = 0$.

(3) Como f es continua en x_0, tenemos que

$$\forall\varepsilon > 0 : \exists\eta > 0 : \forall y \in \mathbb{R}^n : \|y\| < \eta \implies \int_{\|y\|<\eta} \phi_t(y)\,\mathrm{d}y \leqslant \frac{\varepsilon}{4\,\|f\|_\infty}.$$

Entonces, el resultado se sigue igual que en (2) reemplazando x por x_0. ∎

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.